

Theorem 1.1. (The Division Theorem for $\mathbb{Z}$): $a, b \in \mathbb{Z}$ with $b > 0 \exists! q, r \in \mathbb{Z}$ with $0 \leq r < b$ s.t. $a = bq + r$.

Proof. (Existance): Let $R = \{a - bm : m \in \mathbb{Z} \ \& \ a - bm \geq 0\}$, then as: if $a \geq 0$, $a - b \cdot 0 = a \geq 0$ so $a \in R$; if $a < 0$, $0 = a - b \cdot a = a(1 - b) \geq 0$ so $a - b \cdot a \in R$, $R \neq \emptyset$. Thus, take $r = \min(R)$ and so $0 < r = a - bm \in R$ for some $m \in \mathbb{Z}$, choose $q = m \in \mathbb{Z}$. So by construction $a = bq + r$. Note, $r < b$ as were such not true then $r > a - b(q+1) = r - b \in R$ but this contradicts $r = \min(R)$, so we have $0 \geq r < b$. (*Uniqueness*): Suppose $q, q', r, r' \in \mathbb{Z}$ with $0 < r', r < b$ and $a = bq + r = bq' + r'$, also WLOG take $r' \leq r$. So with $\frac{r-r'}{b} \in (-1, 1)$, and $\frac{r-r'}{b} = \frac{a-bq-(a-bq')}{b} = q' - q \in \mathbb{Z}$, we have $\frac{r-r'}{b} = 0$ i.e. $r = r'$. So $q = \frac{a-r}{b} = \frac{a-r'}{b} = q'$. $\square$

Euclidian Algorithm:

Let $a, b \in \mathbb{Z}^+$, apply Th^m 1.1, to give:

$$a = bq_1 + r_1$$

$$b = r_1q_2 + r_2$$

$$r_1 = r_2q_3 + r_3$$

$\vdots$

$$r_{n-1} = r_nq_{n+1},$$

Here, $\gcd(a, b) = r_n$. To prove this relation $G(r_i, r_{i+1}) = G(r_{i+1}, r_{i+2})$, where $G(a, b) = \{n \in \mathbb{Z} : n|a \ \& \ n|b\}$.

Reverse Euclid:

Given a completed algorithm we can obtain $n, m \in \mathbb{Z}$ s.t. $na + mb = \gcd(a, b)$ i.e. taking the end of a Euclidian algothm:

$\vdots$

$$r_{n-3} = r_{n-2}q_{n-1} + r_{n-1}$$

$$r_{n-2} = r_{n-1}q_n + r_n$$

$$r_{n-1} = r_nq_{n+1}$$

we have:

$$(3) \implies r_n = r_{n-2} - r_{n-1}q_n$$

$$(2) \implies r_{n-1} = r_{n-3} - r_{n-2}q_{n-1}$$

$$(4) \text{ and } (5) \implies r_n = r_{n-2} -$$

$$(r_{n-3} - r_{n-2}q_{n-1})q_{n-1}$$

$\vdots$

we continue this ladder up, simplifying each time, (Note: $r_n = \gcd(a, b)$)

Theorem 1.9. Every integer $n \geq 2$ can be written as a product of primes.

Proof. (By strong induction):

(Base case, $n = 2$): Trivially true, $n = 2$ a product of times.

(Inductive Hypothesis): Suppose for $n \in 1, 2, \dots, k$ $n = p_{n,1}p_{n,2} \dots p_{n,m_n}$.

Then for $n = k + 1$, if $k + 1$ is prime we are done, if not then $k + 1 = ab$ where $a, b \in \{2, 3, \dots, k\}$, thus $a = p_{a,1}p_{a,2} \dots p_{a,m_a}$ and $b = p_{b,1}p_{b,2} \dots p_{b,m_b}$ so: $k + 1 = p_{a,1}p_{a,2} \dots p_{a,m_a}p_{b,1}p_{b,2} \dots p_{b,m_b}$. So $\forall 2 < n \in \mathbb{N}$ we are done.

Theorem 1.10. (Euclid's property of primes) $\forall p \in \text{Primes} \ \& \ \forall a, b \in \mathbb{Z}$. $p|ab \implies p|a$ or $p|b$.

Proof. (Case $p|a$): We are done.

(Case $p \nmid a$): Then using the Euclidian algorithm we obtain: $1 = na - mp$ with $n, m \in \mathbb{Z}$. So $p|b \iff p|(na - mp)b \iff p|n(ab) - mbp \iff p|ab$. $\square$

the chain of implications uses a proposition in the notes which states: $p|a \ \& \ p|b \implies p|a + b \ \& \ p|ab$.

Theorem 1.12. (Fundamental Theorem of Arithmetic) $\forall 2 < n \in \mathbb{N}, \exists! p_1, p_2, \dots, p_r \in \text{Primes}$ up to ordering, s.t.: $n = p_1p_2 \dots p_r$.

Proof. The existance is from Th^m 1.9. For uniqueness, suppose $p_1p_2 \dots p_r = q_1q_2 \dots q_s$, then cancel common primes, suppose the result is not 1 i.e. we are left with $p_1p_2 \dots p_n = q_1q_2 \dots q_m$ with $p_i \neq q_j, \forall i \in \{1, \dots, n\}$ and $\forall j \in \{1, \dots, m\}$, after reindexing. Then $p_1|q_1q_2 \dots q_m$ so $\exists j \in \{1, \dots, m\} p_1|q_j$, but q_j is prime so $p_1 = q_j, \nexists$. $\square$

Lemma 1.16. $\forall a, b \in R$: $0a = a0 = 0$; $a(-b) = (-a)b = -(ab)$; $\& \ (-a)(-b) = ab$

Proof. $0a + 0 = 0a = (0 + 0)a = 0a + 0a$ so $0a = 0$. $ab + a(-b) = a(b - b) = a0 = 0 = (a - a)b = ab + (-a)b$ so by the Groups and Geo^m we have $-(ab)$ is the unique element s.t. $ab - (ab) = 0$, $a(-b) = -(ab) = (-a)b$. $(-a)(-b) = -a(-b) = -(-(ab)) = ab$. $\square$

Lemma 1.18. (Subring Test) Let R be a ring and $S \subseteq R$. Then S is a subring of R iff the following hold, $\forall r, s \in S$:

1. $1_R \in S$,
2. $r + s, rs \in S$, and
3. $-r \in S$.

Proof. By 1 and 3 we have $-1_R \in S$ so $0_R = 1_R - 1_R \in S$ by 2. So $(S, +)$ is a group by the subgroup test with $(R, +)$. Commutativity of $+$, associativity of $+$ and $\times$, and distributivity of $\times$ over $+$ are all inherited from R . Note, as they shair operations they shair the identities of the operations, so as, $0_R \neq 1_R$ in R we have $0_S = 0_R \neq 1_R = 1_S$ in S , so (R1)-(R4) are satisfied. $\square$

Lemma 2.2. Suppose $\text{Char}(R) = n > 0$, then: (i) $n \cdot r = 0, \forall r \in R$; $\& \$ (ii) if $m \in \mathbb{Z}^+$ then $m \cdot 1 = 0$ iff $n|m$.

Proof. (i):

$$\begin{aligned} n \cdot r &= \underbrace{r + r + \dots + r}_{n \text{ times}} \\ &= \underbrace{r(1 + 1 + \dots + 1)}_{n \text{ times}} = r0 = 0 \end{aligned}$$

(ii): ($\implies$): Let $m \cdot 1 = 0$ with $m \in \mathbb{Z}^+$, applying the division Th^m with $m = nq + r$ where $q, r \in \mathbb{Z}$ and $0 \leq r < n$ so $0 = m \cdot 1 = (nq + r) \cdot 1 = q(n \cdot 1) + r \cdot 1 = q \cdot 0 + r \cdot 1 = r \cdot 1$, so $r = 0$ as otherwise $n > r \in \mathbb{Z}^+$ and $r \cdot 1 = 0$, a $\nexists$ of n 's defⁿ. (*impliedby*): Let $n|m$ i.e. $\exists q \in \mathbb{Z}$ s.t. $m = qn$, so $m \cdot 1 = qn \cdot 1 = q(n \cdot 1) = q \cdot 0 = 0$. $\square$

Lemma 2.5. If S is a subring of R and R is a domain then S is a domain.

Proof. Let $r, s \in S$ with $rs = 0$, thus $rs = 0$ in R , hence $r = 0$ or $s = 0$. $\square$

Proposition 2.7. Suppose R is a domain. Then $R[X]$ is a domain.

Proof. We use the contrapositive of the definition of a domain [i.e.: R a domain iff $\forall 0 \neq r, s \in R, rs \neq 0$]. Let $0 \neq f, g \in R[X]$, taking the leading coefficients of f $\& \ g$ as $a \neq 0$ and $b \neq 0$, respectively, we the leading coefficient of fg as $ab \neq 0$, hence $fg \neq 0$. $\square$

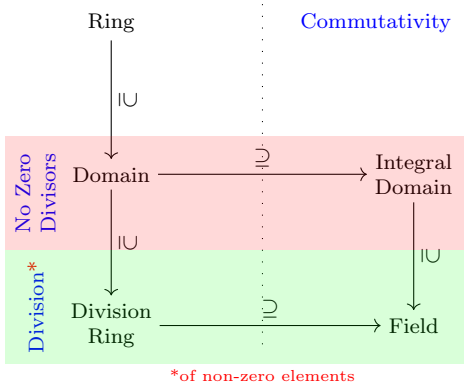
Note: This immediately (by induction on n) yields $R[X_1, \dots, X_n]$ and domain if R is a domain.

Lemma 2.11. If R is a ring and $r \in R$ has both a left and right inverse then these equal and unique.

Proof. (Equal): Choose, $rs = 1 = tr$, then: $s = 1s = trs = t1 = t$.

(Unique right inverse): Let $rs = 1 = rs'$ then $s = 1s = trs = t1 = trs' = 1s' = s'$, where t is choosen s.t. $tr = 0$.

(Unique left inverse): Let $tr = t'r$ then $t = t1 = trs = 1s = t'rs = t'1 = t'$, where s is choosen s.t. $rs = 0$ $\square$



Definition 1.4. $\forall a, b \in \mathbb{Z}$, $\gcd(a, b) = \max\{n \in \mathbb{Z} : n|a \ \& \ n|b\}$, noting $\gcd(0, 0) = 0$ and if $\gcd(a, b) = 1$ then a and b are coprime.

Definition 1.6. $\forall a, b, c \in \mathbb{Z}$ with $n \geq 2$. a is **congruent to b modulo n** iff $n|a - b$, written $a \equiv b \pmod n$. Then define the congruence class of a w.r.t. n as: $[a]_n = \{b \in \mathbb{Z}b \equiv a \pmod n\} = \{b \in \mathbb{Z} : b|m - a\}$

We thus have:

$$\mathbb{Z}_n = \{[a]_n : a \in \mathbb{Z}\} = \{[0]_n, [1]_n, [2]_n, \dots, [n-1]_n\}$$

Note: $b \equiv a \pmod n$ iff $b = a + kn$ iff $[a]_n = \{a + nk : k \in \mathbb{Z}\}$ iff $[a]_n = [a + kn]_n$

Definition 1.14. A **ring** is a set R with two binary operations $+$ $\& \ \times$, defined on R , satisfying the following:

(R1) $(R, +)$ is an abelian group with identity 0.

(R2) $\times$ is associative.

(R3) $\times$ distributes over $+$ i.e.:

$$a(b+c) = ab+ac, \ \& \ (a+b)c = ac+bc, \ \forall a, b, c \in R$$

(R4) The multiplicative identity $1 \in R$ and $1 \neq 0$

Example. $M_n(R)$ is the ring of $n \times n$ matrices, a non-commutative ring. Note: the only ideals of $M_n(R)$ are $M_n(R)$ and $\{0\}$ but this isn't a division ring hence even relaxing commutative in Pro^P 3.22 doesn't yeild the same pro^P with field $\rightarrow$ division ring.

Example. For X a set $\mathcal{P}(X)$ with

$$A + B = (A \cup B) \setminus (A \cap B), \quad AB = A \cap B$$

and $0 = \emptyset$, $-A = A$, $A, B \in \mathcal{P}(X)$.

Definition 1.17. Let R be a ring and $S \subseteq R$. Then S is a **subring** of R if S is a ring w.r.t. the same addition and multiplication as R and contains the multiplicative identity of R , 1_R .

Definition 1.21. let R be a ring. The **ring of polynomials** $R[X] = \{\sum_{i \geq 0} a_i X^i \mid \text{a finite sum}\}$ $a_i \in R$ with addition and multiplication defined as:

$$\sum_{i \geq 0} a_i X^i + \sum_{i \geq 0} b_i X^i = \sum_{i \geq 0} (a_i + b_i) X^i$$

$$\& \ (\sum_{i \geq 0} a_i X^i)(\sum_{i \geq 0} b_i X^i) = \sum_{i \geq 0} (\sum_{j=0}^{n+m} a_j b_{i-j}) X^i$$

$0 = \sum_{i \geq 0} 0X^i \ \& \ 1 = 1X^0 + \sum_{i \geq 1} 0X^i$. Then the **degree** of a polynomial is $i \in \mathbb{N} \cup \{0\}$ s.t. $a_i \neq 0$ and $\deg(0) = -\infty$

More genearely, we have:

$$R[X_1, \dots, X_n] = R[X_1, \dots, X_{n-1}][X_n]$$

$\forall f, g \in R[X], \deg(f + g) \leq \max(\deg(f), \deg(g))$ and $\deg(fg) \leq \deg(f) + \deg(g)$.

Definition 1.24. If R_1 and R_2 are rings then the **Cartesian product** $R_1 \times R_2$ with the $+$ and $\times$ defined element wise is a ring.

Definition 2.1. For $a \in R$ define: for $n \in \mathbb{Z}^+$, $n \cdot a = \underbrace{a + a + \dots + a}_{n \text{ times}}$; for $n = 0$, $n \cdot a = 0 \cdot a = 0$;

$\& \$ for $n \in \mathbb{Z}^-$, $n \cdot a = (-n) \cdot (-a)$.

The **Characteristic**, $\text{char}(R)$, of a ring R is the least positive integer n s.t. $n \cdot 1 = 0$.

Note: $(n + m) \cdot a = n \cdot a + m \cdot a$, $n \cdot (a + b) = n \cdot a + n \cdot b$, $\& \ n \cdot (m \cdot a) = (nm) \cdot a$, all easily verifiable.

Definition 2.3. A non-zero element $r \in R$ is a **zero-divisor** if $\exists s \in R$ a non-zero element s.t. $rs = 0$ or $sr = 0$.

A ring R is a domain if, $\forall r, s \in R$:

$$rs = 0 \implies r = 0 \text{ or } s = 0.$$

i.e. a ring with no zero divisors is a domain and vice versa.

Definition 2.9. A **division ring** is a ring where every non-zero element has a right and left inverse, i.e.: $\forall r \in R, \exists s, t \in R$ s.t. $rs = 1 = tr$ with s, t being the right and left inverses, respectively.

Note: this immediately collapse, due to Le^m 2.11, to just there exists an inverse as $s = t$.

An **invertible** element r is called a **unit**.